5.4 Normed Vector Spaces

Definition 5.4.1 (Valued Field)
*Let K be a field. A **valuation** on K is a function*

$$|\cdot| : K \to \mathbb{R}_{\geq 0}$$

such that for all $x, y \in K$ the following relations are satisfied

a) $|x| = 0 \iff x = 0$
b) $|xy| = |x||y|$
c) $|x + y| \leq |x| + |y|$

*We say that this valuation is **non-archimedean** if it satisfies the ultrametric inequality*

$$|x + y| \leq \max(|x|, |y|)$$

*for all $x, y \in K$. Otherwise it is **archimedean**.*

*We say the pair $(K, |\cdot|)$ is a **valued field**.*

Example 5.4.2
Any subfield of $\mathbb{C}$ (and therefore $\mathbb{R}$) is an archimedean valued field, using the complex absolute value $|\cdot|$ (5.2.2).

Definition 5.4.3 (Normed Vector Space)
Let $(K, |\cdot|)$ be a valued field and X a K-vector space. A norm $\|\cdot\|$ is a function

$$\|\cdot\| : X \to \mathbb{R}_{\geq 0}$$

satisfying the following properties

a) $\|x\| = 0 \iff x = 0$
b) $\|\lambda x\| = |\lambda|\|x\|$
c) $\|x + y\| \leq \|x\| + \|y\|$

*We say the pair $(X, \|\cdot\|)$ is a **normed vector space** over the valued field $(K, |\cdot|)$.*

*A normed vector space which is complete as a metric space is called a **Banach space**.*

A Banach space X which is also an associative algebra satisfying

$$\|x \cdot y\| \leq \|x\|\|y\|$$

*is a **Banach Algebra**. Note a valued field is naturally a Banach Algebra.*

Example 5.4.4
A valued field (e.g. $\mathbb{C}$, $\mathbb{R}$, $\mathbb{Q}$) is a normed vector space over itself.

$\mathbb{C}$ is a normed vector space over $\mathbb{R}$.

For any valued field K the vector space K^n is a normed vector space with the following definition of norm

$$\|v\| := \sqrt{v_1^2 + \cdots + v_n^2}$$

(the triangle inequality requires proof).

Proposition 5.4.5
Let $(X, \|\cdot\|)$ a normed vector space. Then it is naturally a metric space with induced metric

$$d(x, y) := \|x - y\|$$

and in particular a Hausdorff topological space with base the open balls

$$B(x; \varepsilon) := \{y \in X \mid \|y - x\| < \varepsilon\}$$

Furthermore the open balls $B(x; \varepsilon)$ and $B\left(x; \frac{1}{n}\right)$ form a local base at x.

The same statement applies to valued fields and in particular $\mathbb{R}$.

© David Rufino CC BY-NC-ND 4.0

5.4.1 Continuous Functions

Proposition 5.4.6 (Continuous maps of Normed Vector Spaces)
Let $S \subset X$ be a subset of a normed vector space and $f : S \to Y$ be a map to a normed vector space Y and $x \in S$. Then the following are equivalent

a) For all $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x' \in S$, $\|x - x'\| < \delta \implies \|f(x) - f(x')\| < \varepsilon$
b) f is continuous at x with respect to the subspace topology on S

*In this case we say the map f is **continuous** at x.*

Example 5.4.7
For X a normed vector space then $\|\cdot\| : X \to \mathbb{R}$ is evidently continuous.

5.4.2 Product Space

Definition 5.4.8 (Product Norm)
*Let $X_1, \cdots, X_n$ be normed vector spaces over the same valued field K. Then we may define the **product norm** on the vector space $X_1 \times \cdots \times X_n$ by*

$$\|(x_1, \cdots, x_n)\| := \max(\|x_1\|, \cdots, \|x_n\|)$$

Proposition 5.4.9
Let $X_1, \cdots, X_n$ be normed vector spaces over the same valued field K. Then the metric (resp. topology) induced by product norm is the same as product metric (resp. product topology).

Proof. Consider a point $x = (x_1, \cdots, x_n)$ then

$$B(x; \varepsilon) = \{(x_1, \cdots, x_n) \mid \|x_i\| \leq \varepsilon\} = B(x_1; \varepsilon) \times \cdots \times B(x_n; \varepsilon)$$

Therefore the open balls are open in the product topology, and being a base for the norm topology, this is a subset of the product topology (...). To show that the product topology is a subset of the norm topology it is sufficient to show that open sets of the form

$$U := U_1 \times \cdots \times U_n$$

are open in the product norm topology. For given $y \in U$ then by definition $B(y_i; \varepsilon_i) \subset U_i$. Consider $\varepsilon := \min(\varepsilon_1, \cdots, \varepsilon_n)$ and $z \in B(y; \varepsilon)$. Then

$$\|z - y\| < \varepsilon \implies \max_i(\|z_i - y_i\|) < \varepsilon \leq \varepsilon_i \implies z_i \in B(y_i; \varepsilon_i) \implies z \in U$$

in other words $B(y; \varepsilon) \subset U$ as required. ∎

Proposition 5.4.10 (Arithmetic Operations are Continuous)
Let X be a normed vector space over a valued field K. The following maps are continuous

*a) **addition** $+ : X \times X \to X$*
*b) **scalar multiplication** $\cdot : K \times X \to X$*
*c) **inverse scalar** $(-)^{-1} : K \setminus \{0\} \to K$*

Similarly if X is a Banach algebra then the multiplication operator

$$\cdot : X \times X \to K$$

is also continuous.

Proof. By the triangle inequality

$$\|(x_1, x_2) - (y_1, y_2)\| < \varepsilon \implies \|x_1 - y_1\| < \varepsilon \wedge \|x_2 - y_2\| < \varepsilon \implies \|(x_1 + x_2) - (y_1 + y_2)\| < 2\varepsilon$$

Therefore we may apply the criterion given by (5.4.6), and using the product norm.

For scalar multiplication observe

$$\begin{aligned}
\|(\lambda_1, x_1) - (\lambda_2, x_2)\| < \delta &\implies |\lambda_1 - \lambda_2| < \delta \wedge \|x_1 - x_2\| < \delta \\
&\implies \|\lambda_1 x_1 - \lambda_2 x_2\| \leq |\lambda_1| \|x_1 - x_2\| + \|x_2\| |\lambda_1 - \lambda_2| \\
&\implies \|\lambda_1 x_1 - \lambda_2 x_2\| \leq |\lambda_1| \|x_1 - x_2\| + (\|x_1\| + \delta) |\lambda_1 - \lambda_2| \\
&\implies \|\lambda_1 x_1 - \lambda_2 x_2\| \leq \delta^2 + (\|x_1\| + \delta)\delta
\end{aligned}$$

Therefore choose $\delta < \min\left(1, \frac{\varepsilon}{\|x_1\|+1}\right)$ to find $\|\lambda_1 x_1 - \lambda_2 x_2\| < 3\varepsilon$. ∎

© David Rufino CC BY-NC-ND 4.0

5.4.3 Convergent Sequences

Proposition 5.4.11 (Convergent Sequence)
Let X be a normed vector space and (a_n) a sequence in X. Then the following are equivalent

a) $a_n \to \alpha$ in the topological sense (4.1.49)

b) For all $\varepsilon > 0$ there exists $N(\varepsilon) \in \mathbb{N}^+$ such that

$$n \geq N(\varepsilon) \implies \|a_n - \alpha\| < \varepsilon$$

c) The function

$$\begin{aligned} \mathbb{N} \cup \{\infty\} &\to X \\ n &\to a_n \\ \infty &\to \alpha \end{aligned}$$

is continuous with respect to the topology on $\mathbb{N} \cup \{\infty\}$ given by open sets of the form $\{N, N+1, \cdots\} \cup \{\infty\}$ and arbitrary subsets of $\mathbb{N}$.

*In this case we say the sequence (a_n) is **convergent**. If this doesn't hold it is **divergent**.*

Proposition 5.4.12 (Uniqueness of Limits)
Let X be a normed vector space. Then it is Hausdorff and limits are unique.

Proposition 5.4.13 (Continuous Image of Convergent Sequence)
Let X, Y be normed vector spaces, $S \subset X$ and $f : S \to Y$ a map. Then the following are equivalent

a) f is continuous in the topological sense

b) For every convergent sequence $a_n \to \alpha$ we have $f(a_n) \to f(\alpha)$.

Proof. Recall that X is first-countable and so the equivalence follows from (4.1.50) and (4.1.52). ∎

Proposition 5.4.14 (Arithmetic of Convergent Sequences in an NVS)
Let X be a normed vector space and (a_n) and (b_n) be convergent sequences in X. Then

$$\lim_{n\to\infty} (a_n + b_n) = \lim_{n\to\infty} a_n + \lim_{n\to\infty} b_n$$

$$\lim_{n\to\infty} \lambda a_n = \lambda \lim_{n\to\infty} a_n$$

In particular $a_n \to \alpha \iff a_n - \alpha \to 0$.

Proposition 5.4.15 (Arithmetic of Convergent Sequences in a valued field)
Let K be a valued field and (a_n) and (b_n) be convergent sequences. Then

$$\lim_{n\to\infty} a_n b_n = \left(\lim_{n\to\infty} a_n\right)\left(\lim_{n\to\infty} b_n\right)$$

Further if both b_n and $\lim_{n\to\infty} b_n$ is non-zero then

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{\lim_{n\to\infty} a_n}{\lim_{n\to\infty} b_n}$$

Proposition 5.4.16 (Arithmetic of Convergent Sequences in an ordered field)
Let K be an ordered field and (b_n) a sequence of non-negative elements. Then $b_n \to \infty \iff b_n^{-1} \to 0$.

Proposition 5.4.17
Let K be a valued field and $x \in K$ such that $|x| < 1$. Then

$$\lim_{n\to\infty} x^n = 0$$

Proof. We may reduce to the case that $x \in \mathbb{R}$ and $0 < x < 1$. Let $y = \frac{1-x}{x} > 0$, then it is enough to show that $(1+y)^n \to \infty$ by (5.4.16). However from (5.1.30) $(1+y)^n \geq 1 + ny$, and the result follows from Archimedean property (...). ∎

Definition 5.4.18 (Cauchy Sequence)
*Let X be a normed vector space and $(a_n)_{n\in\mathbb{N}}$ a sequence in X. We say (a_n) is a **cauchy sequence** if for all $\varepsilon > 0$ there exists $N(\varepsilon)$ such that*

$$n, m \geq N(\varepsilon) \implies \|a_n - a_m\| < \varepsilon$$

© David Rufino CC BY-NC-ND 4.0

*Every **convergent sequence** is **cauchy**. If the converse holds then we say X is **complete**.*

5.4.4 Finite-Dimensional Normed Spaces

Definition 5.4.19 (Equivalent Norms)
*Let X be a vector space over a valued field K. Consider two norms $\|\cdot\|_1$ and $\|\cdot\|_2$. We say that they are **equivalent** if there exists constants $c_1, c_2 > 0$ such that*

$$c_1\|x\|_1 \leq \|x\|_2 \leq c_2\|x\|_1$$

We may prove easily that this is an equivalence relation on norms.

Proposition 5.4.20
Let X be a vector space over a valued field K. Suppose that $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent norms for X. Then the following derived quantities are the same

- *Induced topology*
- *Convergent sequences*
- *Cauchy sequences*

Proposition 5.4.21 (Supremum Norm)
Let X be a finite-dimensional vector space with basis $(e_1, \cdots, e_n)$. Then the supremum norm defined by

$$\left\|\sum_{i=1}^n \lambda_i e_i\right\|_\infty := \max(|\lambda_1|, \cdots, |\lambda_n|)$$

is well-defined (it is the product norm on $X \cong K \times \cdots \times K$). If K is complete then so is X.

Proposition 5.4.22 (Equivalence of Norms)
Let X be a non-zero finite-dimensional vector space over a complete valued field K. Then every norm is equivalent to the supremum norm (and therefore complete).

Proof. Fix a basis $(e_1, \cdots, e_n)$. Then for any norm $\|\cdot\|$ we have by the triangle inequality

$$\left\|\sum_{i=1}^n \lambda_i e_i\right\| \leq \sum_{i=1}^n |\lambda_i|\|e_i\| \leq \left(\sum_{i=1}^n \|e_i\|\right)\left\|\sum_{i=1}^n \lambda_i e_i\right\|_\infty$$

The reverse inequality we prove by induction. The case $n = 1$ we may take $c_1 = \|e_1\|$. Suppose $n > 1$ and there is no such constant C. Then there are vectors $v_k \in X$ such that

$$\|v_k\| < \frac{1}{k}\|v_k\|_\infty$$

Suppose

$$v_k = \sum_{i=1}^n \lambda_{ki} e_i$$

Consider the sets of integers

$$S_i := \left\{k \in \mathbb{N} \;\middle|\; \|v_k\|_\infty = |\lambda_{ki}|\right\} \quad i = 1 \cdots n$$

Then at least one of $S_1, \cdots, S_n$ is infinite, since the union is $\mathbb{N}$. Without loss of generality we assume it is S_n.

Therefore by passing to a subsequence, and scaling, we may assume that $\|v_k\|_\infty = \lambda_{kn} = 1$ and $\|v_k\| < \frac{1}{k}$. Define the vector $w_k := v_k - e_n$. Then by definition $w_k \to e_n$ with respect to $\|\cdot\|$.

Let W be the $(n-1)$-dimensional subspace spanned by $\{e_1, \cdots, e_{n-1}\}$. We prove also that $w_k \to w \in W$ with respect to $\|\cdot\|$, which contradicts uniqueness of limits. Evidently w_k is a cauchy sequence with respect to $\|\cdot\|$ restricted to W. By induction $\|\cdot\|_\infty$ is equivalent to $\|\cdot\|$ on W, and therefore w_k is cauchy with respect to $\|\cdot\|_\infty$. As W is complete (5.4.21) we see that $w_k \to w$ with respect to $\|\cdot\|_\infty$, and therefore with respect to $\|\cdot\|$. ∎

Corollary 5.4.23
Let X be a finite dimensional normed vector space with basis $\{x_1, \cdots, x_n\}$. The projection map

$$\pi_i : X \to \langle x_i \rangle \xrightarrow{\sim} K$$

© David Rufino CC BY-NC-ND 4.0

given by $\pi_i(x) = x_i^\vee(x)x_i$ is a continuous, open map.

5.4.5 Convergent Series

Definition 5.4.24 (Series)
Let X be a normed vector space and (a_n) a sequence in X. The formal expression

$$\sum_{n=1}^{\infty} a_n$$

*is called the **series** associated to this sequence. For such a sequence define the **partial sum** sequence as follows*

$$s_n := \sum_{k=1}^{n} a_k$$

*If the partial sum sequence converges then we say that the series **converges** and write*

$$\sum_{n=1}^{\infty} a_n := \lim_{n\to\infty} s_n$$

If the series

$$\sum_{n=1}^{\infty} \|a_n\|$$

*converges then we say the original series **converges absolutely**.*

The choice of terminology is justified in the case that X is complete.

Proposition 5.4.25 (Absolute Convergence $\implies$ Convergence)
Let X be a complete normed vector space and $\sum_{n=0}^{\infty} a_n$ a series which converges absolutely. Then the series converges in the usual sense.

Proof. The partial sums of the series $\sum_{n=1}^{\infty} \|a_n\|$ are cauchy by (5.4.18). By the triangle inequality we may demonstrate that the partial sums of the series $\sum_{n=1}^{\infty} a_n$ are cauchy. Then by completeness of X the partial sums must also converge. ∎

Proposition 5.4.26
Let X be a normed vector space and $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ two convergent sequences. Then the following series are convergent with limits

$$\sum_{n=1}^{\infty}(a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$$

$$\sum_{n=1}^{\infty} \lambda a_n = \lambda \sum_{n=1}^{\infty} a_n$$

where $\lambda \in K$.

Proposition 5.4.27
Let X be a normed vector space and $\sum_{n=1}^{\infty} a_n$ a convergent series. Then $\|a_n\| \to 0$.

Proof. Let $\alpha := \sum_{n=1}^{\infty} a_n$. Then by definition there exists N such that $n \geq N \implies \left\|\sum_{k=1}^{n} a_k - \alpha\right\| < \varepsilon$. As this holds for $n+1$ we may use the triangle inequality to find that $n \geq N+1 \implies \|a_n\| < 2\varepsilon$. As ε was arbitrary we find $\|a_n\| \to 0$ as required. ∎

Proposition 5.4.28 (Comparison Test)
Let X be a complete normed vector space. Suppose (a_n) is a sequence in X and (c_n) is a sequence in $\mathbb{R}$ such that

a) The series $\sum_{n=0}^{\infty} c_n$ converges
b) $\|a_n\| \leq c_n$ for $n \geq N_0$

then the series $\sum_{n=0}^{\infty} a_n$ converges absolutely.

Proof. By assumption *b)* (c_n) is non-negative, and by (5.4.29) the sequence of partial sums $\sum_{k=0}^{n} c_k$ is bounded. therefore so is the sequence $\sum_{k=0}^{n} \|a_k\|$, and by the same result the series $\sum_{n=0}^{\infty} \|a_n\|$ converges, in other words $\sum_{n=0}^{\infty} a_n$ converges absolutely. ∎

© David Rufino CC BY-NC-ND 4.0

Proposition 5.4.29 (Series of Positive Reals)
Let (a_n) be a sequence of non-negative real numbers. Then the following are equivalent
a) The sequence of partial sums $A_n := \sum_{k=0}^{n} a_k$ is bounded
b) The series $\sum_{n=0}^{\infty} a_n$ converges (absolutely)

Proof. By (5.1.23) we always have $A_n \to \sup A_n \in \mathbb{R}^\sharp$, and the limit is finite iff A_n is bounded. ∎

Proposition 5.4.30 (Geometric Series)
Suppose $0 \leq x < 1$ is a real number, then

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

and in particular the partial sums are bounded by $\frac{1}{1-x}$.

Proof. It's straightforward to show by induction that the partial sums are given explicitly by

$$\sum_{n=0}^{N} x^n = \frac{1-x^N}{1-x} \ .$$

Then the result follows from (5.4.17). ∎

Proposition 5.4.31 (Root Test)
Let X be a normed vector space. Given a series $\sum_{n=0}^{\infty} a_n$, define $\alpha := \limsup_n \sqrt[n]{\|a_n\|}$ Then
a) if $\alpha < 1$ then the series converges absolutely
b) if $\alpha > 1$ then the series diverges

Proof.
a) Define $\alpha := \limsup_n \sqrt[n]{\|a_n\|}$ and $\beta := \frac{1+\alpha}{2} < 1$. Then by definition there is an integer N such that

$$n \geq N \implies \sqrt[n]{\|a_n\|} \leq \alpha + \frac{1-\alpha}{2} = \beta \implies \|a_n\|^n \leq \beta^n$$

In particular for $n \geq N$

$$\sum_{k=0}^{n} \|a_k\| \leq \sum_{n=0}^{N-1} \|a_k\| + \sum_{k=N}^{n} \|a_k\| \leq \sum_{k=0}^{N-1} \|a_k\| + \sum_{k=N}^{n} \beta^k \leq \sum_{k=0}^{N-1} \|a_k\| + \sum_{k=0}^{\infty} \beta^k$$

where the last step follows from by (5.4.30). We conclude from (5.4.29) that the series converges absolutely.
b) By (5.1.35) there exists a subsequence a_{n_k} such that $\sqrt[n_k]{\left\|a_{n_k}\right\|} \to \alpha > 1$. Therefore we may find a subsequence such that $\sqrt[n_k]{\left\|a_{n_k}\right\|} \geq \frac{1+\alpha}{2} > 1 \implies \left\|a_{n_k}\right\| > 1$. This shows that $\left\|a_{n_k}\right\| \not\to 0$, whence $\|a_n\| \not\to 0$ by (...), and the series diverges by (5.4.27).

∎

Proposition 5.4.32 (Ratio Test)
Let K be a valued field and consider a series $\sum_{n=1}^{\infty} a_n$. If

$$\limsup_n \left|\frac{a_{n+1}}{a_n}\right| < 1$$

then the series is convergent.

Proof. Let $\alpha := \limsup \left|\frac{a_{n+1}}{a_n}\right|$ and $\beta = \frac{1+\alpha}{2}$. Then by assumption $\beta < 1$, and there exists N such that $n \geq N \implies \left|\frac{a_{n+1}}{a_n}\right| \leq \beta \implies |a_{N+k}| \leq |a_N|\beta^k$. We then see by Comparison Test (5.4.28) that the series is convergent. ∎

Proposition 5.4.33 (Cauchy Product of Series)
Let X be a complete valued field and $\sum_{n=1}^{\infty} a_n$, $\sum_{n=1}^{\infty} b_n$, $\sum_{n=1}^{\infty} c_n$ series such that
a) one of $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are absolutely convergent
b) $c_n := \sum_{k=0}^{n} a_k b_{n-k} = \sum_{k=0}^{n} a_{n-k} b_k$

then $\sum_{n=0}^{\infty} c_n$ is absolutely convergent and

$$\sum_{n=0}^{\infty} c_n = \left(\sum_{n=1}^{\infty} a_n\right)\left(\sum_{n=1}^{\infty} b_n\right)$$

© David Rufino CC BY-NC-ND 4.0

Proof. Denote the partial sums by A_n, B_n and C_n, and the limits by A, B respectively. Then evidently

$$C_n = \sum_{i=0}^{n} a_{n-i} B_i$$

therefore

$$C_n = \sum_{i=0}^{n} a_{n-i}(B_i - B) + A_n B$$

Then it's sufficient to show that the first term tends to 0. Observe that $\beta_n := B_n - B \to 0$. **TODO** ∎

5.4.6 Function Spaces

Definition 5.4.34 (Uniform Convergence)
*Let S be a metric space and Y a normed vector space. Suppose $\{f_n : S \to Y\}$ is a sequence of functions and $f : S \to Y$ is a function. We say that $f_n \to f$ **converges uniformly** if*

$$\forall \varepsilon > 0 \exists N \text{ s.t. } n \geq N \implies \|f_n(x) - f(x)\| < \varepsilon \forall x \in S$$

Note this implies, but is strictly stronger than, pointwise convergence that is $f_n(x) \to f(x)$ for all $x \in S$, as N may be chosen independently of x.

Proposition 5.4.35 (Uniform Limit of Continuous Functions is Continuous)
Let S be a metric space and Y a normed vector space. Suppose that $\{f_n : S \to Y\}$ is a sequence of continuous functions converging uniformly to a limit $f : S \to Y$. Then f is continuous.

Proof. Fix $\varepsilon > 0$. There exists N such that

$$n \geq N \implies \|f_n(x) - f(x)\| < \varepsilon \forall x \in S$$

Fix $x \in S$ then there exists $\delta > 0$ such that

$$\|y - x\| < \delta \implies \|f_n(y) - f_n(x)\| < \varepsilon$$

Therefore

$$\|y - x\| < \delta \implies \|f(x) - f(y)\| \leq \|f(x) - f_n(x)\| + \|f_n(x) - f_n(y)\| + \|f(y) - f_n(y)\| < 3\varepsilon$$

Consequently f is also continuous. ∎

Definition 5.4.36 (Bounded Function Space)
*Let S be a metric space and Y a normed vector space. Define the vector space of **bounded functions** by*

$$\mathcal{B}(S, Y) := \{f : S \to Y \mid \exists C > 0 \text{ s.t. } \|f(x)\| \leq C \text{ for all } x \in S\}$$

*This is a normed vector space under the **sup norm** given by*

$$\|f\|_\infty := \sup_{x \in S} \|f(x)\|$$

Evidently convergence in this norm is equivalent to uniform convergence.

Proposition 5.4.37 (Bounded Functions are Complete)
Suppose Y is a Banach Space, then $\mathcal{B}(S, Y)$ is also a Banach Space.

Proof. Let $\{f_n\}$ be a cauchy sequence of maps $S \to Y$ with respect to the sup norm. Then for every $\varepsilon > 0$ there exists N such that

$$n, m \geq N \implies \|f_n - f_m\| < \varepsilon$$

In particular for every $x \in S$ we have $\{f_n(x)\}$ is cauchy. As Y is complete it is a convergent sequence, for which we denote the limit by $f(x)$. For every $x \in S$ we may also choose some $m \geq N$ such that

$$\|f_m(x) - f(x)\| < \varepsilon$$

then

$$\|f(x) - f_n(x)\| \leq \|f(x) - f_m(x)\| + \|f_m(x) - f_n(x)\| < 2\varepsilon$$

© David Rufino CC BY-NC-ND 4.0

As this inequality is independent of m we find that $\|f - f_n\| \to 0$ as required. We may also show easily that it is bounded. ∎

5.4.7 Continuous Linear Maps

Proposition 5.4.38 (Continuous Linear Maps)
Let $f : X \to Y$ be a linear map of normed vector spaces. Then the following are equivalent

a) f is continuous

b) There exists a constant $C > 0$ such that

$$\|f(x)\| \leq C\|x\|$$

c) f is bounded on the unit sphere

$$\sup\{\|f(x)\|_Y \mid \|x\|_X = 1\} < \infty$$

When X is finite-dimensional then every linear map is continuous.

Proof. Suppose b) holds then f is even uniformly continuous because

$$\|x - y\| < \frac{\varepsilon}{C} \implies \|f(x) - f(y)\| < \varepsilon$$

Conversely given a) there exists δ such that

$$\|x\| \leq \delta \implies \|f(x)\| < 1$$

Therefore

$$\left\| f\left(\frac{\delta x}{\|x\|}\right) \right\| < 1 \implies \|f(x)\| < \frac{1}{\delta}\|x\|$$

and we may take $C = \frac{1}{\delta}$. Given b) evidently $\|f\| \leq C < \infty$, and conversely $\|f(x)\| \leq \|f\|\|x\|$. ∎

Proposition 5.4.39
Let X, Y be normed vector spaces. Denote by $L(X, Y)$ the set of continuous linear maps. It is a normed vector space, with norm given by

$$\|\theta\| := \sup\{\|\theta(x)\|_Y \mid \|x\|_X = 1\}$$

Furthermore if Y is complete, so is $L(X, Y)$.

Proof. Evidently $\|\lambda\theta\| = |\lambda|\|\theta\|$ and $\|\theta_1 + \theta_2\| \leq \|\theta_1\| + \|\theta_2\| < \infty$. Let $\{\theta_n\}$ be a cauchy sequence in $L(X, Y)$. Given $x \in X$ then

$$\|\theta_n(x) - \theta_m(x)\| \leq \|\theta_n - \theta_m\|\|x\|$$

whence $\{\theta_n(x)\}$ is a cauchy sequence, and therefore converges to $\theta(x)$. By uniqueness of limits we see that θ is linear. Furthermore by the reverse triangle inequality (...)

$$|\|\theta_n\| - \|\theta_m\|| \leq \|\theta_n - \theta_m\|$$

in otherwords $\|\theta_n\|$ is a cauchy sequence and converges to some limit C. Suppose that $\|x\| = 1$ then

$$\|\theta(x)\| \leq \|\theta(x) - \theta_n(x)\| + \|\theta_n(x)\| \leq \|\theta(x) - \theta_n(x)\| + \|\theta_n\|$$

Choose N_1 such that $n \geq N_1 \implies \|\theta_n\| \leq C + \varepsilon$ and N_2 such that $n \geq N_2 \implies \|\theta(x) - \theta_n(x)\| < \varepsilon$. As ε was arbitrary we see that $\|\theta\| \leq C$ and in particular is finite. ∎

Proposition 5.4.40 (Finite-Dimensional Linear Maps are continuous)
Suppose that $\theta : X \to Y$ is a linear map of normed vector spaces and X is finite-dimensional. Then θ is continuous.

Proof. Let $\{e_1, \cdots, e_n\}$ be a basis. By (5.4.22) we know that $\|\cdot\|_X$ is equivalent to the infinity norm $\|\cdot\|_\infty$. therefore it is sufficient to show that θ is continuous with respect to this norm. However

$$\left\| \theta\left(\sum_{i=1}^n \lambda_i e_i\right) \right\| \leq \sum_{i=1}^n |\lambda_i|\|\theta(e_i)\| \leq n\left\|\sum_{i=1}^n \lambda_i e_i\right\|_\infty \max_i \|\theta(e_i)\|$$

∎

© David Rufino CC BY-NC-ND 4.0

Proposition 5.4.41 (Continuous Multilinear Maps)
Let $\psi : X_1 \times \cdots \times X_p \to Z$ be a multilinear map of normed vector spaces (using the product norm). Then the following are equivalent

a) ψ is continuous

b) There exists a constant $C > 0$ such that

$$\|\psi(x_1, \cdots, x_p)\| \leq C\|x_1\| \cdots \|x_p\|$$

c) The following quantity is finite

$$\sup\{\|\psi(x_1, \cdots, x_p)\| \mid \|x_1\| = \cdots = \|x_p\| = 1\} < \infty$$

In this case $L(X_1, \cdots, X_p; Z)$ is a normed vector space with norm given by expression in c). Furthermore if Z is complete so is $L(X_1, \cdots, X_p; Z)$.

Finally if $X_1, \cdots X_p$ are finite-dimensional then every multilinear map is continuous.

Proof. **a) $\implies$ b)** Suppose there is no such constant C, then there exists $(x_n^1, \cdots, x_n^p) \in X_1 \times \cdots \times X_p$ such that

$$\|\psi(x_n^1, \cdots, x_n^p)\| > n^p \|x_n\| \|y_n\|$$

Define $\hat{x}_n^i := \frac{x_n^i}{n\|x_n\|}$. Then $\|(\hat{x}_n^1, \cdots, \hat{x}_n^p)\| = \frac{1}{n} \to 0$ but $B(0, \cdots, 0) = 0$ by linearity and $B(\hat{x}_n^1, \cdots, \hat{x}_n^p) > 1$, which contradicts continuity of limits (...).

b) $\implies$ a) Observe by linearity

$$\begin{aligned}\psi(x_1, \cdots, x_p) - \psi(y_1, \cdots, y_p) &= \psi(x_1 - y_1, x_2, \cdots, x_p) + \psi(y_1, x_2 - y_2, x_3, \cdots, x_p) + \cdots + \psi(y_1, \cdots, y_{p-1}, x_p - y_p) \\ &\leq C\|x_1 - y_1\|\|x_2\| \cdots \|x_p\| + C\|x_2 - y_2\|\|y_1\|\|x_3\| \cdots \|x_p\| + \cdots \\ &+ C\|x_p - y_p\|\|y_1\| \cdots \|y_{p-1}\|\end{aligned}$$

Fix any $\varepsilon > 0$ then it is enough to show that

$$\|x_i - y_i\| < \min(\|y_i\|, \varepsilon) \implies \|\psi(x_1, \cdots, x_p) - \psi(y_1, \cdots, y_p)\| < M\varepsilon$$

for some positive constant M depending only on y. For this implies that $\|x_i\| < 2\|y_i\|$ so that

$$\|\psi(x_1, \cdots, x_p) - \psi(y_1, \cdots, y_p)\| \leq \varepsilon \sum_{i=1}^{p} 2C\|y_1\| \cdots \widehat{\|y_i\|} \cdots \|y_p\|$$

as required.

b) $\iff$ c) Straightforward

The supremum is evidently a norm, and the verification of completeness follows as before. ∎

Definition 5.4.42 (Space of Continuous Multilinear Maps)
Let $X_1, \cdots, X_p$, Y be normed vector spaces then let $L(X_1, \cdots, X_p; Y)$ denote the K-vector space of all continuous multilinear maps.

Denote by $L^p(X, Y) := L(X^p; Y)$ the vector space of continuous multilinear maps on the p-fold product of X.

Proposition 5.4.43 (Alternating Multilinear Maps)
*Let $L_a(X_1, \cdots, X_p; Y)$ denote the subspace of continuous linear maps $L(X_1, \cdots, X_p; Y)$ which are **alternating** (3.5.5). Then this is a closed subset. In particular when Y is complete so is $L_a(X_1, \cdots, X_p; Y)$.*

As before define $L_a^p(X, Y) := L_a(X^p; Y)$.

© David Rufino CC BY-NC-ND 4.0